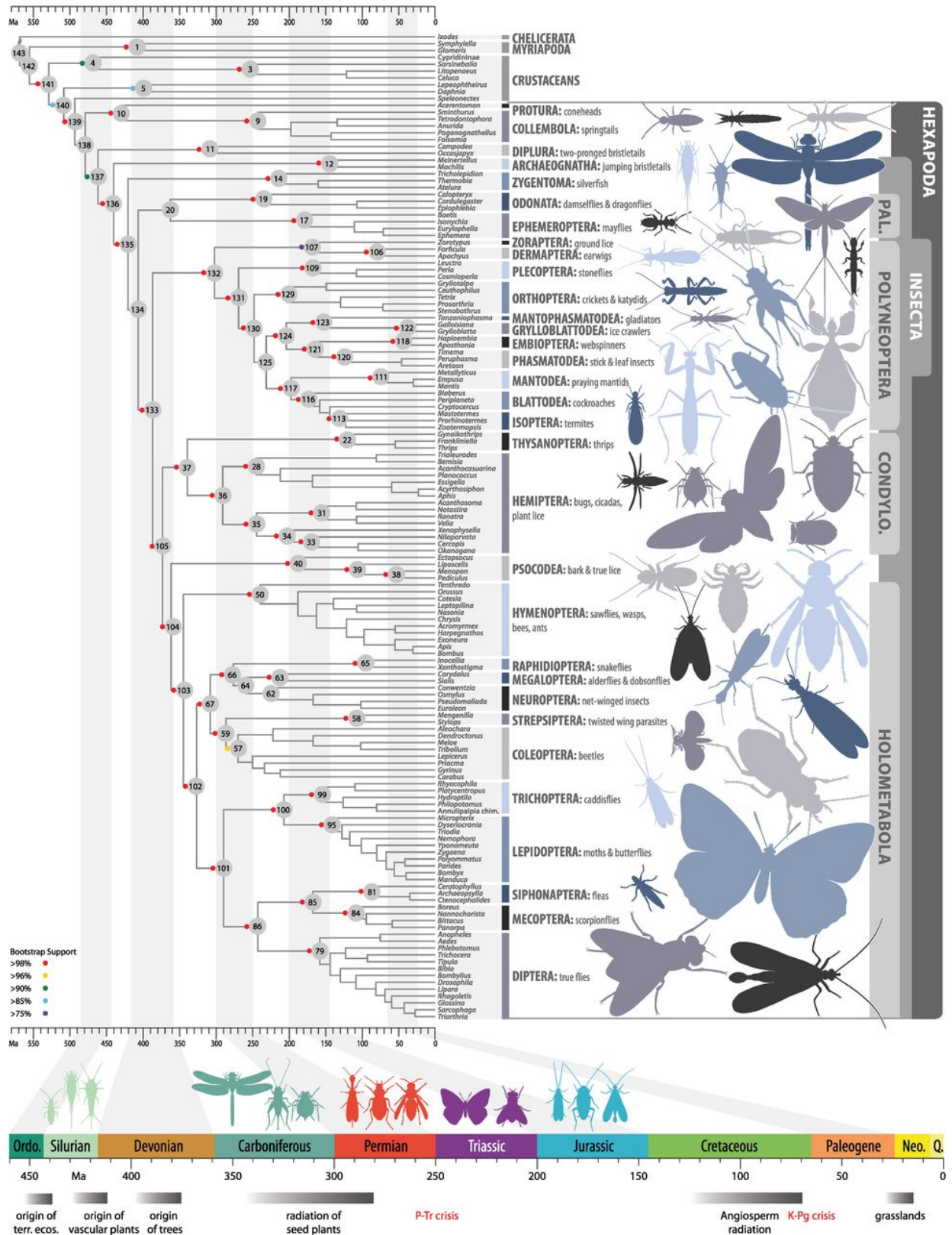


HEXAPOD ORDERS AND FAMILIES



From Misof et al. 2014. Phylogenomics resolves the timing and pattern of insect evolution

1. **PROTURA** (Gr.,protos, first + oura, tail)
2. **COLLEMBOLA** (Gr.,kolla, glue + embolon, peg,wedge)
3. **DIPLURA** (Gr., diploos,double + oura, tail)
 - a. Japygidae
 - b. Campodeidae
4. **ARCHAEOGNATHA** (Jumping Bristle tails; = Microcoryphia)
 - a. Machilidae
5. **ZYGENTOMA** (= *Thysanura sensu stricta*; Gr.,thysanos, tassel + oura, tail) Silverfish
6. **EPHEMEROPTERA** (Gr.,ephemeros, lasting but a day)
7. **ODONATA** (Gr., odontos,tooth + ata, characterized by)
 - Suborder Anisoptera
 - a. Gomphidae
 - b. Libellulidae
 - c. Aeschnidae
 - Suborder Zygoptera
 - a. Calopterygidae
 - b. Coenagrionidae
 - c. Lestidae
8. **PLECOPTERA** (Gr., plekein, to twist + pteron, wing)
9. **EMBIIDINA** (EMBIOPTERA) (Gr.,embios, lively + pteron, wing)
10. **ORTHOPTERA** (Gr., orthos, straight + pteros, winged)
 - a. Tetrigidae
 - b. Acrididae
 - c. Tridactylidae
 - d. Tettigoniidae
 - e. Gryllacrididae
 - f. Gryllidae
 - g. Gryllotalpidae
 - h. Stenopelmatidae
11. **PHASMIDA** (PHASMATODEA)
12. **GRYLLOBLATOIDEA** (GRYLLOBLATTA, NOTOPTERA)
13. **DERMAPTERA** (Gr.,derma, skin + pteron, wing)
14. **ZORAPTERA** (Gr.,zoros, pure, + apterygos, wingless)
15. **MANTODEA**
16. **BLATTODEA** (BLATTARIA)
 - a. Blattidae (oriental, American, and other cockroaches)
 - b. Blatellidae (German, Wood and other cockroaches)
 - c. Blaberidae (Giant Cockroaches)
 - d. **ISOPTERA** (Gr.,isos,equal + pteron, wing)
17. **THYSANOPTERA** (Gr.,thysanos, tassel + pteron,wing)
 - a. Phlaeothripidae
 - b. Thripidae
18. **PSOCODEA**
 - a. **PTHIRAPTERA**
 - b. **PSOCOPTERA** (Gr.,psoco,rub,small + pteron,wing)
19. **HEMIPTERA** (=HETEROPTERA)
 - a. Aphididae
 - b. Coccidae (Scale insects)
 - c. Cicadidae (Cicadas)
 - d. Cercopidae (Spittlebugs)
 - e. Cicadellidae (Leafhopper)
 - f. Fulgoridae
 - g. Gerridae (Water striders)
 - h. Nepidae (Water Scorpions)
 - i. Belostomatidae (Giant water bugs)
 - j. Gelastocoridae (Toad bugs)
 - k. Corixidae (Water boatmen)
 - l. Naucoridae (Creeping water bugs)
 - m. Notonectidae (Backswimmers)
 - n. Cimicidae (Bed Bugs)

- o. Nabidae (Damsel Bugs)
- p. Miridae (Plant bugs)
- q. Reduviidae (Assassin Bugs)
- r. Berytidae (Stilt Bugs)
- s. Lygaeidae (Seed bugs)

- t. Coreidae (Leaf footed bugs)
- u. Rhopalidae (Scentless plant bugs)
- v. Scutellaridae (Shield bugs)
- w. Pentatomidae (Stink bugs)
- 20. **MEGALOPTERA:**
 - a. Corydalidae (Fishflies, dobsonflies)
 - b. Sialidae (Alderflies)
- 21. **RAPHIDIOPTERA**
- 22. **NEUROPTERA**
 - a. Mantispidae (Mantidflies)
 - b. Hemerobiidae (Brown Lacewings)
 - c. Chrysopidae (Green Lacewings)
 - d. Myrmeleontidae (Antlions)
 - e. Ascalaphidae (Owlflies)
- 23. **COLEOPTERA**
 - a. Carabidae (Ground Beetles, including tiger beetles)
 - b. Dytiscidae (Predaceous Diving Beetles)
 - c. Gyrinidae (Whirligig Beetles)
 - d. Silphidae (Carrion Beetles)
 - e. Staphylinidae (Rove Beetles)
 - f. Hydrophilidae (Water Scavenger Beetles)
 - g. Histeridae (Hister Beetles)
 - h. Lucanidae (Stag Beetles)
 - i. Passalidae (Bess/Bessie Beetles/Bugs)
 - j. Scarabaeidae (Scarab Beetles)
 - k. Buprestidae (Metallic Wood-boring Beetles)
 - l. Dryopidae (Long-toed Water Beetles)
 - m. Elmidae (Riffle Beetles)
 - n. Psephenidae (Water-penny Beetles)
 - o. Elateridae (Click Beetles)
 - p. Lycidae (Net-winged Beetles)
 - q. Phengodidae (Glowworms)
 - r. Lampyridae (Lightning Bugs or Fireflies)
 - s. Cantharidae (Soldier Beetles)
 - t. Dermestidae
 - u. Ptinidae (Spider Beetles)
 - v. Cleridae (Checkered Beetles)
 - w. Melyridae (Soft-winged Flower Beetles)
 - x. Nitidulidae (Sap Beetles)
 - y. Cucujidae (Flat Bark Beetles)
 - z. Erotylidae (Pleasing Fungus Beetles)
 - aa. Coccinellidae (Ladybird Beetles)
 - bb. Mordellidae (Tumbling Flower Beetles)
 - cc. Tenebrionidae (Darkling Beetles)
 - dd. Meloidae (Blister Beetles)
 - ee. Pyrochroidae (Fire-colored Beetles)
 - ff. Cerambycidae (Longhorns)
 - gg. Bruchidae (Seed Beetles)
 - hh. Chrysomelidae (Leaf Beetles)
 - ii. Brentidae (Straight-snouted Weevils / Primitive Weevils)
 - jj. Curculionidae (Snout Beetles)
 - kk. Scolytidae (Bark and Ambrosia Beetles)
- 24. **HYMENOPTERA**
 - Suborder Symphyta
 - a. Xyelidae
 - b. Pamphilida
 - c. Cimbicidae

- d. Tenthredinidae
- e. Cephidae
- f. Siricidae

Suborder Apocrita

- a. Evaniidae (ensign wasps)
- b. Braconidae
- c. Ichneumonidae
- d. Chalcididae
- e. Cynipidae
- f. Chrysididae (Cuckoo wasps)
- g. Sphecidae (thread waisted wasps)
- h. Colletidae
- i. Halictidae (sweat bees)
- j. Andrenidae
- k. Megachilidae
- l. Anthophoridae
- m. Apidae (Honey bees, Bumblebees)
- n. Mutillidae (Velvet Ants)
- o. Pompilidae (spider wasps)
- p. Scoliidae
- q. Vespidae
- r. Formicidae

27. **TRICHOPTERA**

- a. Hydropsychidae
- b. Hydroptilidae
- c. Rhyacophilidae
- d. Brachycentridae
- e. Limnephilidae
- f. Phryganeidae
- g. Leptoceridae

28. **LEPIDOPTERA** (Gr.,lepidos,scale + pteron, wing)

- a. Micropterygidae
- b. Sesiidae
- c. Cossidae
- d. Tortricidae
- e. Pterophoridae
- f. Pyralidae
- g. Hesperiidae
- h. Papilionidae
- i. Pieridae
- j. Lycaenidae
- k. Nymphalidae
- l. Satyridae
- m. Danaidae
- n. Geometridae
- o. Saturniidae
- p. Sphingidae
- q. Arctiidae
- r. Noctuidae

29. **MECOPTERA** (Gr.,mecos,length = pteron, wing)

- a. Boreidae (Snow Scorpionflies)
- b. Bittacidae (Hangingflies)
- c. Panorpidae (Scorpionflies)
- d. Panorpididae
- e. Meropeidae

30. **SIPHONAPTERA**

31. **DIPTERA**

- a. Culicidae (mosquitoes)
- b. Tabanidae (horse flies / deer flies)
- c. Simuliidae (black flies)
- d. Psychodidae (moth flies)
- e. Ceratopogonidae (punkies, no-see-ums)
- f. Muscidae (House flies)
- g. Cecidomyiidae (gall midges)
- h. Tephritidae (fruit flies)
- i. Agromyzidae
- j. Anthomyiidae
- k. Drosophilidae (pomace flies)
- l. Tipulidae (crane flies)
- m. Calliphoridae (blow flies)
- n. Chironomidae (midges)
- o. Sarcophagidae (flesh flies)
- p. Asilidae (robber flies)
- q. Bombyliidae (bee flies)
- r. Empididae (dance flies)
- s. Syrphidae (flower flies)
- t. Tachinidae
- u. Sciomyzidae (marsh flies)
- v. Oestridae (bot flies / warble flies)
- w. Hippoboscidae (louse flies)